Computer Vision

Homework 5: Geometry, Light and Learning on Paper

Hieu Nguyen

In the spirit of Assignment 5 of CS 585 (Spring 2024), paper-and-pencil exercises only. Goes with Chapters 2 to 5 of the notes.

1. **Thin lens.** A camera has a 35 mm lens. How far from the lens must the sensor be to focus on an object 1.5 m away? With the aperture at $f/2.8$, what is the blur circle diameter of a point 3 m away? Derive the general formula you use.

2. **Vanishing points.** Show that the vanishing points of all directions parallel to the ground plane lie on a single line in the image, and find that line for a camera tilted down by an angle ϕ with focal length f.

3. **Focal length.** The vanishing points of two perpendicular horizontal directions in a 1280×720 photo are at $(-410, 300)$ and $(1950, 330)$. Assuming the principal point is the image centre, estimate the focal length in pixels and the horizontal field of view.

4. **Stereo.** A stereo rig has $f = 800$ pixels and a baseline of 10 cm, and disparities can be measured to ± 0.25 pixels. Up to what distance is the depth known to within 5%?

5. **Optical flow.** Derive the optical flow constraint equation from brightness constancy, explain the aperture problem in terms of the matrix $A^\top A$ of the Lucas–Kanade method, and compute its eigenvalues for a window containing a single straight edge.

6. **Kalman filter.** For the scalar update with prediction variance P and measurement variance R, show that the variance after the update satisfies $1/P_{\text{new}} = 1/P + 1/R$. For a random walk $x_k = x_{k-1} + w_k$ with $\operatorname{Var}(w_k) = Q$, find the steady-state variance and gain.

7. **Epipolar geometry.** Show that the essential matrix $E = [t]_\times R$ has rank 2, that $E^\top t = 0$ means the epipole in the first image is the image of the second camera's centre, and that $x_2^\top E x_1 = 0$ for every pair of matching normalized points.

8. **Photometric stereo.** Three lights have directions $(0, 0, 1)$, $(0.6, 0, 0.8)$ and $(0, 0.6, 0.8)$. A Lambertian pixel has brightness 0.8, 0.68 and 0.32 under them. Find its albedo and unit normal.

9. **GANs.** Derive the optimal discriminator D^* for a fixed generator and show that the value of the game becomes $-\log 4 + 2\,\mathrm{JS}(p_{\text{data}} \,\|\, p_g)$.

10. **Attention.** If the entries of a query and a key are independent with mean 0 and variance 1, show that their dot product has variance d, and explain what goes wrong with the softmax if we don't divide by $\sqrt{d}$.

11. **Absolute orientation.** Show that the best translation in the absolute orientation problem maps one centroid onto the other, and that the best rotation maximizes $\sum_i b'_i \cdot R\, a'_i$.